Chapter 1 The atom

EXTRA WORKSHEET 2:Half-life and carbon dating

SCIENCE INQUIRY SKILLS **VERBAL/LINGUISTIC, LOGICAL/MATHEMATICAL, VISUAL/SPATIAL**

The difference between a stable isotope and a radioisotope is that a radioisotope may undergo a nuclear reaction at any moment, whereas a stable isotope will not.

It is impossible to know exactly when a particular atom of a radioisotope will undergo a nuclear reaction. Only the probability of the nuclear reaction occurring can be known. This probability can be expressed in terms of the radioisotope's half-life.

Half-life

The half-life of a radioisotope is the time it takes for half of the atoms in any group to undergo a nuclear reaction. It cannot be known specifically which of the atoms will undergo a reaction, but it can be known that approximately half of them will react within one half-life.

For example, if you have 1000 atoms of a radioisotope with a half-life of 5 days, then 500 of the atoms should have reacted after 5 days. The half-life for the remaining 500 atoms is still 5 days, so after another 5 days the number of unreacted atoms will halve again, leaving 250 unreacted atoms. After another 5 days you will have 125 unreacted atoms, and so on.

The percentage of atoms of a radioisotope that remain unreacted after a certain time can be calculated using the equation:

$$\text{Percentage of atoms remaining} = 100 \times \frac{1}{2^{\frac{\text{time}}{\text{half-life}}}}$$

Carbon dating

Scientists can use the half-life equation to calculate the age of old artefacts or fossils in a process known as carbon dating. To do this, scientists measure the amount of the radioisotope carbon-14 in the artefacts. The half-life of carbon-14 is 5730 years, so the percentage of carbon-14 remaining in an artefact changes over time according to the equation:

$$\text{Percentage of atoms remaining} = 100 \times \frac{1}{2^{\frac{\text{time (in years)}}{5730}}}$$

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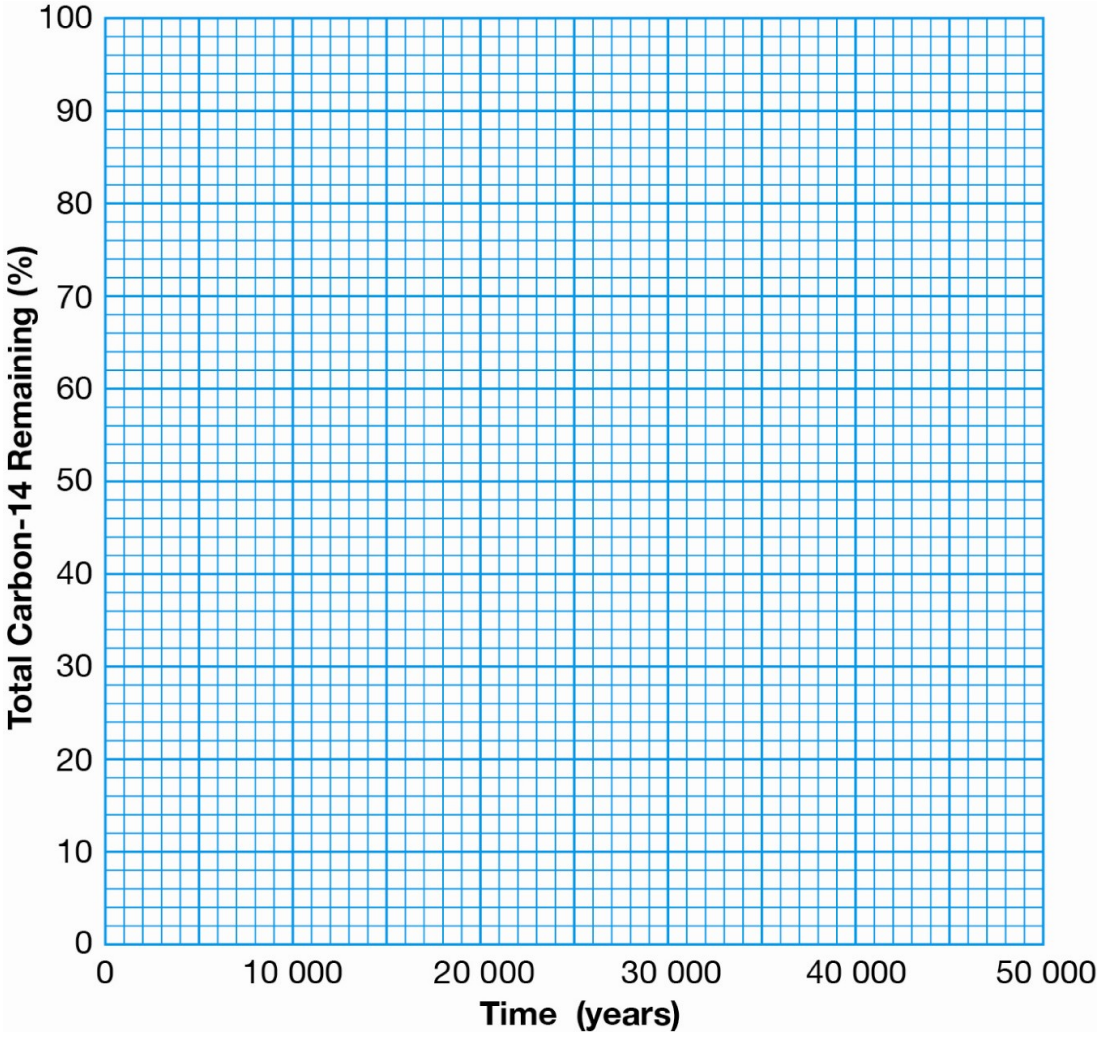
Living things constantly replace their carbon atoms, so the proportion of carbon-14 in a new object is always the same. By comparing the amount of carbon-14 in an old artefact with the amount in similar materials today, scientists can calculate the artefact's age.

1 Use the equation for the half-life of carbon-14 to **complete** the tables below.

Time (years)	0	5000	10 000	15 000	20 000	25 000
Carbon-14 remaining (%)	100	54.6	29.8	16.3	8.9	4.8

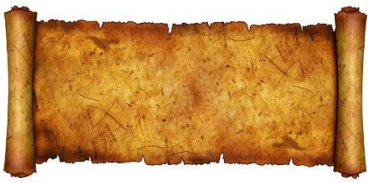
Time (years)	30 000	35 000	40 000	45 000	50 000
Carbon-14 remaining (%)	2.6	1.4	0.8	0.4	0.2

2 **On the axes below, construct a graph of the percentage of carbon-14 as a function of time.**

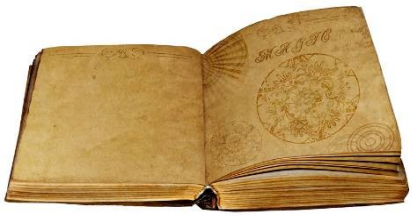


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3 Use your graph or the half-life equation to **calculate** the percentage of carbon-14 that would remain in the following artefacts.

a Papyrus scroll, 4000 years old  ___ %	**b** Tree stump, 28 000 years old  ___ %
c Skull, 17 000 years old  ___ %	**d** Hatchet, 2000 years old  ___ %

4 Use your graph to **determine** the age of the following artefacts to the nearest 500 years.

a Book, 90% of original carbon-14  _____ years	**b** Mallet, 2% of original carbon-14  _____ years
c Wooden spoon, 20% of original carbon-14  _____ years	**d** Feather, 44% of original carbon-14  _____ years

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